# Module 3: ICU Mortality Prediction

**Problem Statement**: To create a mortality prediction model that uses ICU EHR data to predict patient survival.

- Traditional scoring systems are weighted according to parameters that may *not achieve a good prediction sensitivity and specificity*.

## Goal

This module analyzes ICU patients and how their history of **PCa (Prostate Cancer)**, related comorbidities, their vitals, input fluids, procedures affects ICU mortality.

## Clinical Database

**MIMIC - IV** - https://physionet.org/content/mimiciv/2.2/

- The Medical Information Mart for Intensive Care (MIMIC)-IV database provided critical care data for over 40,000 patients admitted to intensive care units at the Beth Israel Deaconess Medical Center (BIDMC).
- MIMIC-IV is sourced from two in-hospital database systems: a custom hospital wide EHR and an ICU specific clinical information system.

# Module 3: ICU Mortality Prediction - Methodology

1. **Dataset preparation**

   The MIMIC-IV version 2.2 is an MIT Licensed Clinical Database, hence access is granted only after review from PhysioNet and credentialed training from CITI and MIT affiliated data handling courses:
   - Conflicts of Interest
   - Data or Specimens Only Research.

   The clinical database consisted of **2 modules**: the icu (9 datasets) and hospital (22 datasets).

   The final dataset consists of **26** features and **1 target** variable.

```
icustays.head()
```

| | subject_id | hadm_id | stay_id | first_careunit | last_careunit | admittime | los | dischtime | deathtime | gender | ... |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 10002430 | 26295318 | 38392119 | Coronary Care Unit (CCU) | Coronary Care Unit (CCU) | 13-06-2129 00:00 | 2.922593 | 24-06-2129 16:01 | NaN | M | ... |
| 1 | 10002443 | 21329021 | 35044219 | Coronary Care Unit (CCU) | Coronary Care Unit (CCU) | 17-10-2183 23:20 | 2.750729 | 20-10-2183 18:47 | NaN | M | ... |

# Module 3: ICU Mortality Prediction - Methodology

```
icustays.head()
```

| | subject_id | hadm_id | stay_id | first_careunit | last_careunit | admittime | los | dischtime | deathtime | gender | ... |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 10002430 | 26295318 | 38392119 | Coronary Care Unit (CCU) | Coronary Care Unit (CCU) | 13-06-2129 00:00 | 2.922593 | 24-06-2129 16:01 | NaN | M | ... |
| 1 | 10002443 | 21329021 | 35044219 | Coronary Care Unit (CCU) | Coronary Care Unit (CCU) | 17-10-2183 23:20 | 2.750729 | 20-10-2183 18:47 | NaN | M | ... |

1. **Dataset preparation**

   Data processing followed these extensive steps:

- Filter *only ICU patients and their related information*.
  - Features included: subject_id, hadm_id, stay_id, first_careunit, last_careunit, admittime, dischtime, deathtime, gender, admission_type, admission_location, language, marital_status, race.
- *Calculating target variable* based on deathtime and expire_status.
  - Patients are assigned labels in categories of **No death** (if they survived that stay_id), **ICU Death** (if death occured in ICU) and **Normal Death** (if death occured after being discharged from ICU).

- Using the diagnoses dataset, ICU patients with a *history of prostate cancer* were identified, and every patient's history of *common comorbidities* of PCa like hypertension, heart failure, diabetes, obesity, chronic obstructive pulmonary disorder, chronic kidney disease were analyzed.

- *Procedure events* like the hours of dialysis and ventilation (invasive and non-invasive) were also added.

- *Inputs of medications* like the amounts of Dextrose 5%, Fentanyl, Gastric Meds, NaCl 0.9%, Norepinephrine, Potassium Chloride administered to patients were documented.

# Module 3: ICU Mortality Prediction - Methodology

## 2. Model Preparation and Selection

Two different models were prepared: Tree-based and Weight-based.

- Tree-based: An XGBoost model was developed with the following parameters:
  - objective: multi:softmax , random_state=42, tree_method=gpu_hist, num_classes=3
  - GridSearch for Cross Validation was performed on learning_rate, gamma, max_depth, n_estimators, min_child_weight to select the most optimal parameters for model selection
  - Cross validation fits 5 folds for each of 243 candidates, for a total of 1215 fits.
  - The best parameters are as follows:
    - 'gamma': 0.5
    - 'learning_rate': 0.1
    - 'max_depth': 5
    - 'min_child_weight': 1
    - 'n_estimators': 75

```
Fitting 5 folds for each of 243 candidates, totalling 1215 fits
Out[20]:
GridSearchCV
estimator: XGBClassifier
XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None, feature_types=None,
              gamma=None, grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=None, max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=None, max_leaves=None,
              min_child_weight=None, missing=nan, monotone_constraints=None,
              multi_strategy=None, n_estimators=None, n_jobs=None,
```

# Module 3: ICU Mortality Prediction - Methodology

## 2. <u>Model Preparation</u>

Two different models were prepared: **Tree-based** and **Weight-based**.

- <u>Weight-based</u>: A custom ANN model i.e. a feed-forward neural network was implemented using PyTorch's nn.Module.
    - Has **three FC layers**, where FC1 takes an input of size 26 and the final layer FC4 takes the output from the third layer as input (size 4) and outputs a size of 3. This is the **output layer** of the network.
    - The **forward method** defines the forward pass of the network. The input x is passed through each layer with a ReLU activation function applied after the first three layers.
    - After the final layer, **a softmax function is applied to the output** i.e. in this case which is a multi-class classification problem that gets the probabilities of each class.

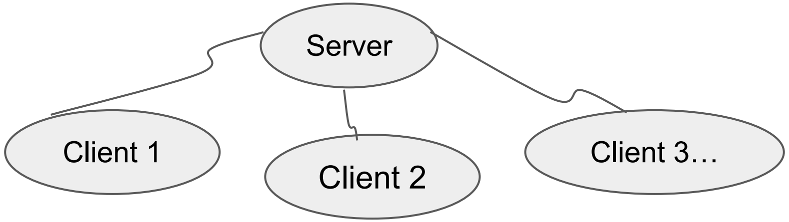
    - An estimator was defined with K-fold stratification, n_splits=5, and run for 10 epochs and a batch_size=32.

This result <u>*was also higher than the baseline*</u>. The metrics of both the models were cross-validated to prevent overfitting.

# Module 3: ICU Mortality Prediction - Methodology

## 3. Federated Learning Implementation

- Weight-based federated learning was selected and the **Flower + pyTorch** federated framework was utilized.
- Three modules were designed - the **server, the client and the model**.
  - The **model** module consists of a script that's used for training and testing the neural network model.
  - The **client** module connects with the Flower server and loads a model, trains it on local data, and communicates the model parameters back to the server.
  - The **server** module connects with various clients and aggregates the model parameters received from the clients. It sends updated parameters back to the clients after using strategies – here, two federated learning algorithms were tested: FedAvg and FedProx.



# Module 3: ICU Mortality Prediction

## Result

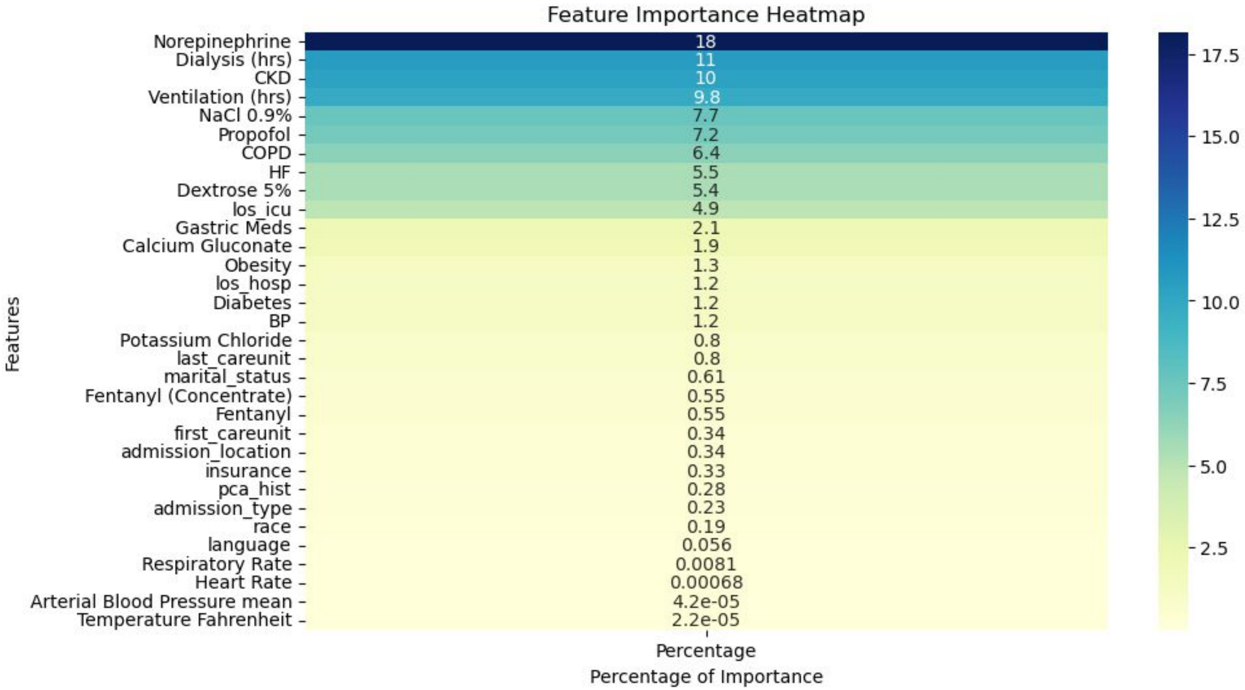



**Fig: ICU Mortality Dataset Feature Importance Heatmap**

## Module 3: ICU Mortality Prediction

### Result

| Strategy | Hyperparameters | Server Accuracy | F1-Score |
|---|---|---|---|
| FedAvg | - | 95.3 | 94 |
| FedProx | Proximal Mu = 0.2 | 94.69 | 93.3 |

### Analysis

- Baseline for the F1-Score for ICU mortality prediction among PCa is 0.82.
- The tree-based XGBoost model gave a test accuracy of 91.638% with an F1-score of 0.91. This cross validated result is higher than the existing baseline of ICU Mortality for PCa using the MIMIC-IV dataset.
- The ANN-model achieved a mean Cross-Validation score of 0.89 and the F1 Score of the model on the test dataset was 0.854, in the centralized architecture.
- Our study identifies significant medication and comorbidities and the impact of a PCa history on an ICU patient's mortality.
- A history of prostate cancer may not wield significant influence over ICU mortality as much as having a comorbidity of Chronic Kidney Disease or the duration of ventilation and dialysis.
- ICU mortality prediction model using FedAvg, achieves accuracy of 95.3 and an F1-score of 94.